



Course Code	: SWE311
Course Name	: Mobile Application System Design
Lecturer	: Nurul Amera binti Muhamad Nazmi
Academic Session	: 2026/04
Assessment Title	: Assignment 1
Submission Due Date	: 10 May 2026

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1. Task 1: Mobile App Design Concept

1.1 App Concept, Purpose, and Target Users

Soluna is a mobile health application designed as a digital ritual journal. Its name "Soluna" comes from "sun" and "moon", symbolising the balance between daily energy and quiet reflection. It integrates journaling, manifestation, emotional tracking, habit tracking and personal reflection into a pleasant mobile experience. Unlike ordinary note-and-note or diary applications, Soluna is committed to helping users establish mindful daily routine. Users can write daily intentions, reflect on emotions, create manifestation goals, record gratitude, and review emotional progress through data insights.

The main purpose of Soluna is to create a warm digital space so that users can slow down, understand their feelings, and develop positive personal habits. The app aims to help users improve emotional awareness and promote self-growth in a simple and inspiring way. For example, the application will not directly ask the user "How are you feeling?", but uses more gentle and reflective wording, such as "What makes you feel lighter today?". This makes the user experience more personalised, calm and non-judgmental.

The target users are mainly young people and adults aged 18 to 40. They are interested in self-improvement, mindfulness, emotional health, journaling and personal growth. These users may include students, young people in the workplace, or anyone who wants to relieve stress, reflect on daily emotions, or develop healthier living habits. Given that the target users may use the app at quiet times such as early morning or before sleeping, the app is designed to create a peaceful, simple and easy-to-use experience.

1.2 Modern Mobile Design Trends

Soluna follows the following modern mobile design trends. The reasons behind these design choices will be explained in Section 1.3.

Modern Design Trend	How It Appears in App
① Minimalist and focused layout	Each screen has one clear purpose and avoids unnecessary visual clutter.
② Card-based interface design	Rounded cards organize intentions, journal prompts, manifest items, insights, and settings.

Modern Design Trend	How It Appears in App
③ Bottom tab navigation	The main sections are fixed at the bottom: Home, Journal, Manifest, Insights, and Profile.
④ Soft visual hierarchy	Large headings, section titles, accent buttons, and muted secondary text guide attention clearly.
⑤ Modern wellness color palette	Warm cream, lavender, peach, sage green, and dusty pink create a calm wellness feeling.
⑥ Rounded components and soft shadows	Buttons, chips, cards, and navigation bars use rounded corners and subtle shadows.
⑦ Personalization and emotional interaction	Goal selection, mood selection, guided prompts, progress tracking, and insights make the app user-centred.
⑧ Consistent visual identity	Typography, colours, icons, spacing, card style, and celestial-botanical details are repeated across screens.

1.3 Justification of the Design Decisions

Soluna's design decision aims to support the application as the core positioning of a quiet and meaningful digital ritual journal. Every visual and structural choice is closely related to the needs of target users and the nature of the app.

Design Decision	Reason for the Choice	User Experience Value
① Minimalist and focused layout	Soluna is intended for users who may open the app during quiet personal moments, such as before sleeping, after studying, or during emotional reflection. A crowded interface may make the experience feel more stressful and less personal. Research on mobile UI design shows that poor interface design and unnecessary complexity can	Users can understand what to do on each screen without spending too much effort interpreting the interface. This supports a calmer and more comfortable reflection experience.

Design Decision	Reason for the Choice	User Experience Value
	increase users' extraneous cognitive load, especially on small mobile screens (Faudzi et al., 2024). Therefore, each Soluna screen is designed around one main purpose. For example, the "Journal Writing Page" mainly focusses on writing prompt and input areas, while the "Insights Page" organises emotional data into clear sections.	
② Card-based interface design	The application contains a variety of short and relevant content, such as daily intentions, journal tips, emotional states, manifest goals, insights and personal information. Card-style design is very suitable because the card can classify relevant information into a clear visual container, making the content easier to browse (Laubheimer, 2016). In Soluna, rounded cards are used to divide sections such as "habit progress" and "manifest board".	The interface is easier to browse and the layout is more orderly. Users can quickly identify different functional areas without being overwhelmed by information overload.
③ Bottom tab navigation	Soluna has five main sections that users may need to access frequently: Home, Journal, Manifest, Insights, and Profile. Bottom tab navigation is suitable for mobile apps because it allows users to move between major app sections easily on handheld screens. Apple's Human Interface Guidelines describe tab bars as a way to switch between different app sections, while Material Design also presents bottom navigation bars as a common pattern for moving between top-level destinations (Apple, 2026 ; Google, n.d.).	Users can freely switch between the main functions of the application without confusion. The fixed bottom navigation bar can also give users a sense of control, because they can return to the main section at any time.
④ Soft visual hierarchy	Soluna draws users' attention through headings, section titles, accent buttons, muted secondary text, and spacing. However, its hierarchical design strives to remain soft and avoid using strong contrasts or	Important actions stand out clearly, but the app still feels calm and emotionally safe. Users can focus on the actions without being distracted by too

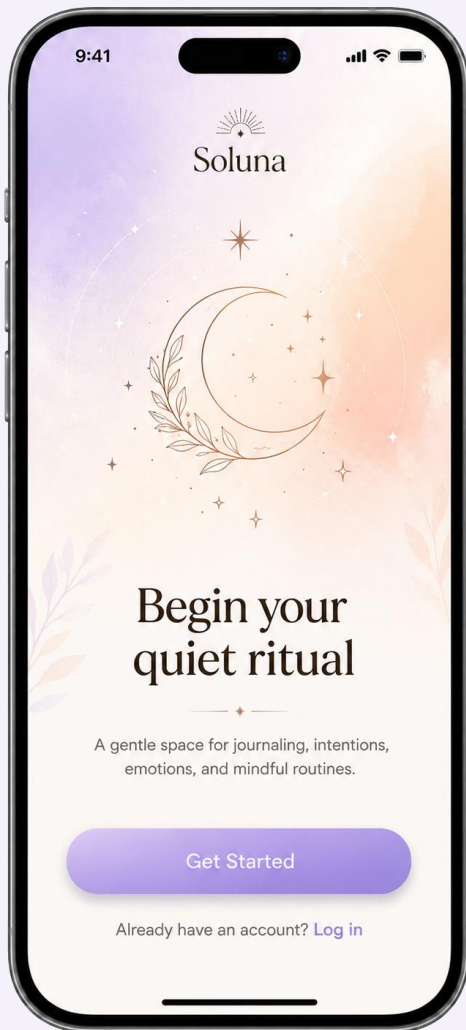
Design Decision	Reason for the Choice	User Experience Value
	radical visual effects. This follows the principle of "progressive disclosure": users should first see the most important options, and the secondary details are only displayed when needed (Nielsen, 2006). For example, the Home Page presents daily actions first, while deeper features such as creating a new manifest are placed inside related screens.	many competing elements.
⑤ Modern wellness color palette	The warm cream background, lavender, peach, sage green, and dusty pink accents were selected to support the app's wellness identity. This is not only an aesthetic choice. Colour psychology research suggests that colour can influence users' emotions, attention, and perception of a digital interface (Fialkowski & Schofield, 2024). Softer and lighter colours are more suitable for Soluna because the app focuses on reflection, comfort, and self-growth. Harsh colours such as strong red or neon tones are avoided because they may feel too urgent or intense for a journaling and wellbeing app.	The colour palette helps create a peaceful, friendly, and emotionally supportive atmosphere. It makes the app feel suitable for personal reflection, especially during quiet moments such as morning planning or night journaling.
⑥ Rounded components and soft shadows	Rounded cards, buttons, chips, input fields, and navigation bars are used to create a softer and more approachable visual style. Subtle shadows and thin borders help separate interactive components from the background without making the interface look heavy. Card interface guidance also explains that shadows, borders, and background contrast can help users recognize grouped and clickable elements (Laubheimer, 2016).	Interactive elements become easier to recognize while still maintaining a gentle visual tone. The interface feels friendly, modern, and less formal than sharp rectangular layouts.
⑦ Personalization and emotional interaction	Soluna includes goal selection, mood selection, guided prompts, manifest creation, habit progress, and emotional insights. These features make the	Users feel that the app supports their own emotional routines and personal growth. This can

Design Decision	Reason for the Choice	User Experience Value
	experience feel more personal instead of generic. Research on mobile health and behaviour-change apps suggests that personalization can improve engagement because the app experience becomes more connected to the user's own profile, needs, and behaviour (Gosetto et al., 2023). The wording is also designed to be gentle, such as using reflective prompts instead of direct or judgmental questions.	encourage repeated use because the app feels meaningful and relevant to the user's life.
⑧ Consistent visual identity	The same colour palette, typography style, icon language, card design, spacing system, and celestial-botanical details are repeated across the screens. Consistency is important because it helps users build familiarity when moving through the app. In Soluna, warm backgrounds, elegant headings, soft cards, rounded buttons, moon/star/leaf details, and the bottom navigation are used repeatedly to create a unified design system.	Users will feel more at ease, because each interface is familiar and coherent. In addition, the application also looks more professional, because each interface seems to belong to a complete product, rather than its own independent design.

In general, these design decisions are not only based on visual beauty, but also closely related to the purpose of the application, target users, and the principle of mobile useability. The minimalist layout reduces the cognitive load, the card system optimises the content grouping, the bottom tab navigation supports users to switch frequently between core functions, the soft colour scheme creates a quiet emotional tone, and the personalised function makes the user experience more meaningful.

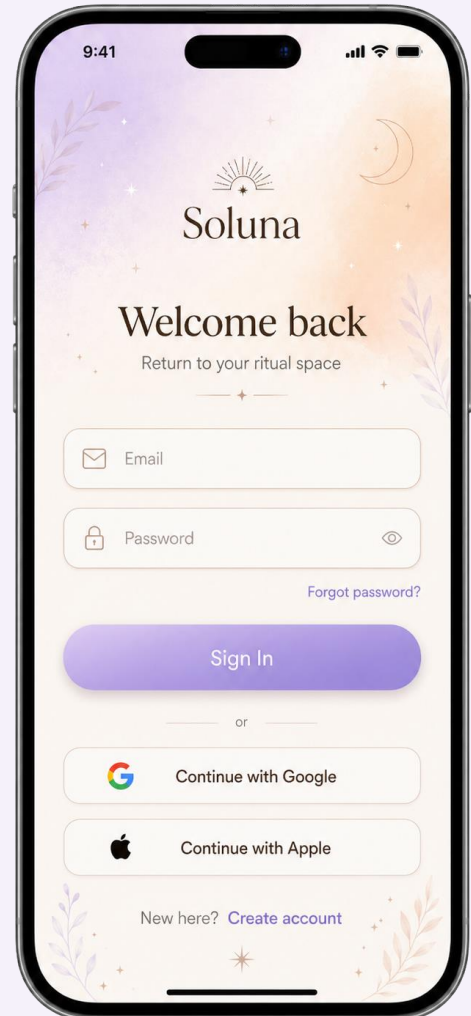
2. Task 2: Mobile UI Screen Design

2.1 UI Screen Gallery



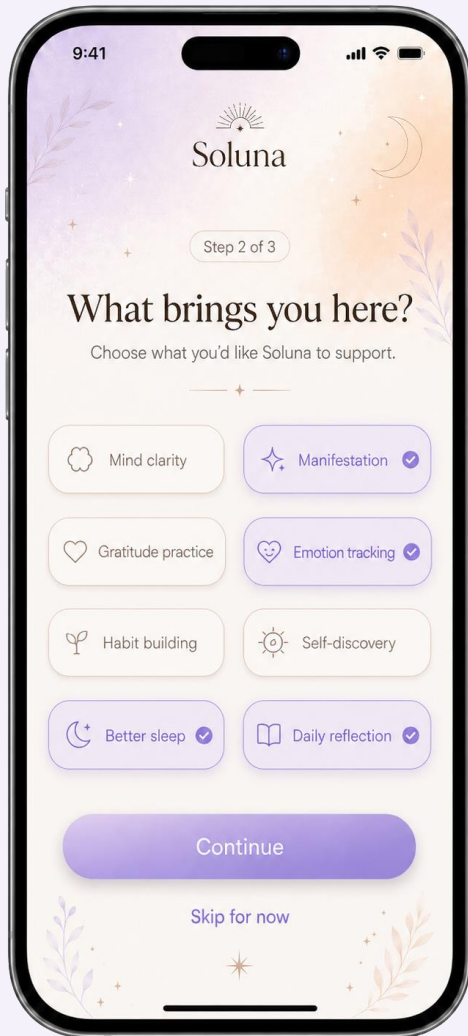
Screen 1. Welcome Page

Brief function: Introduces the application.



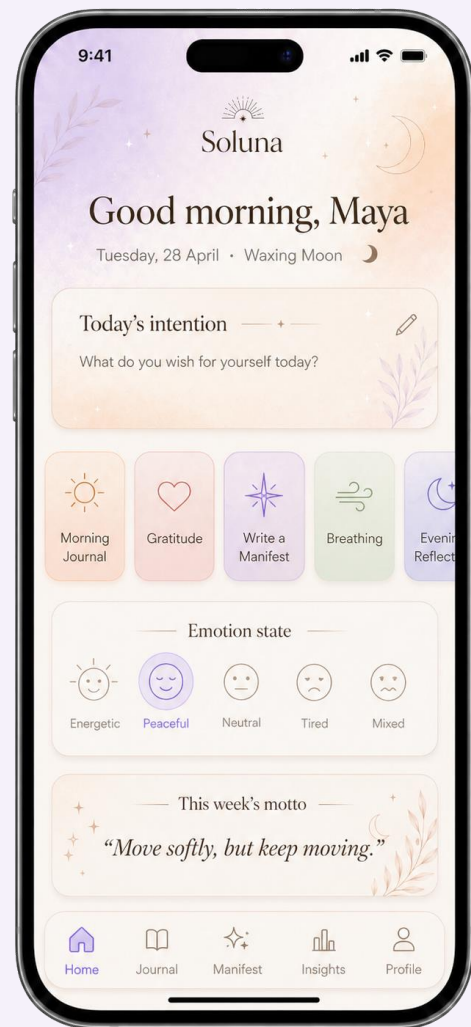
Screen 2. Login Page

Brief function: Allows users to sign in or create an account.



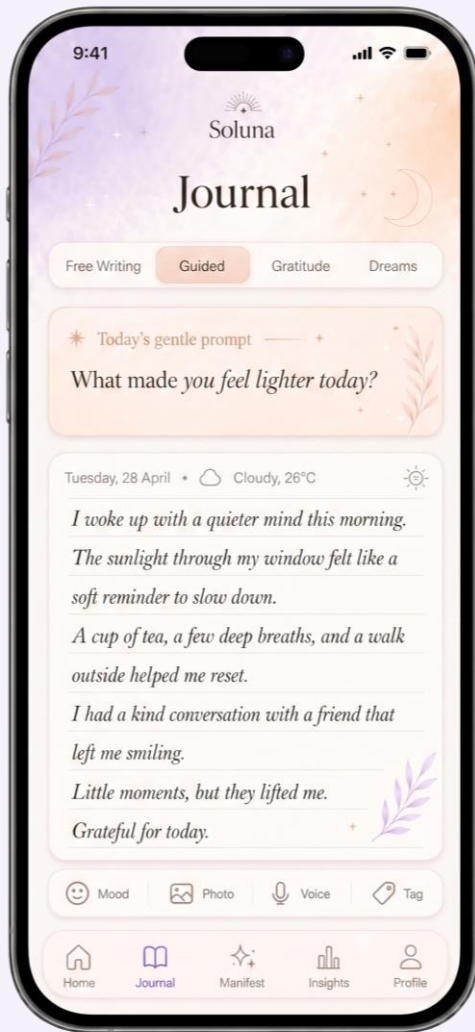
Screen 3. Personal Setup Page

Brief function: Collects user goals to personalize the app experience.



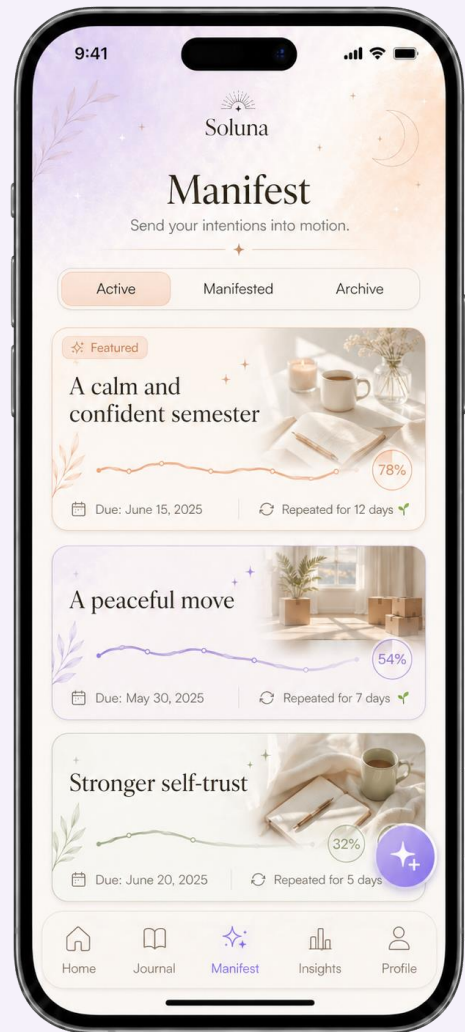
Screen 4. Home Page

Brief function: Presents daily intention, ritual entry points, emotion state, and bottom navigation.



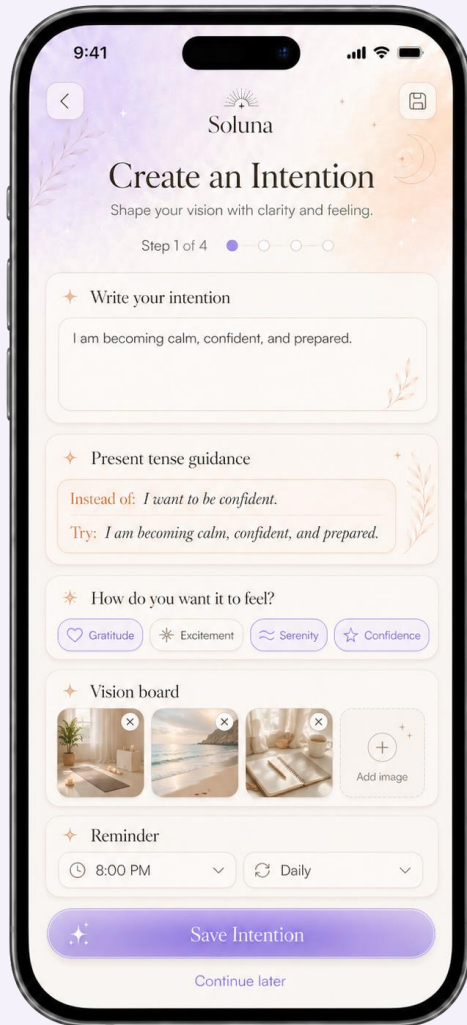
Screen 5. Journal Writing Page

Brief function: Supports guided writing.



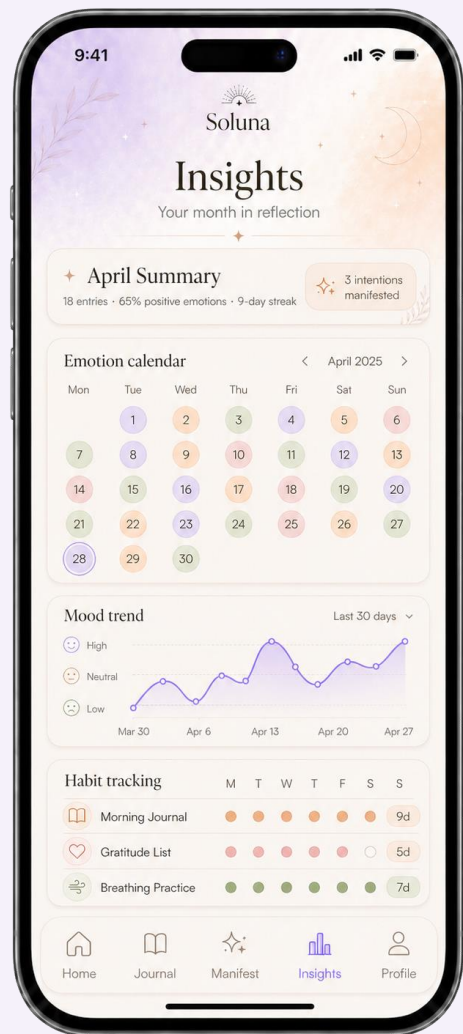
Screen 6. Manifest Board Page

Brief function: Displays active intentions with progress, due dates, and repetition counts.



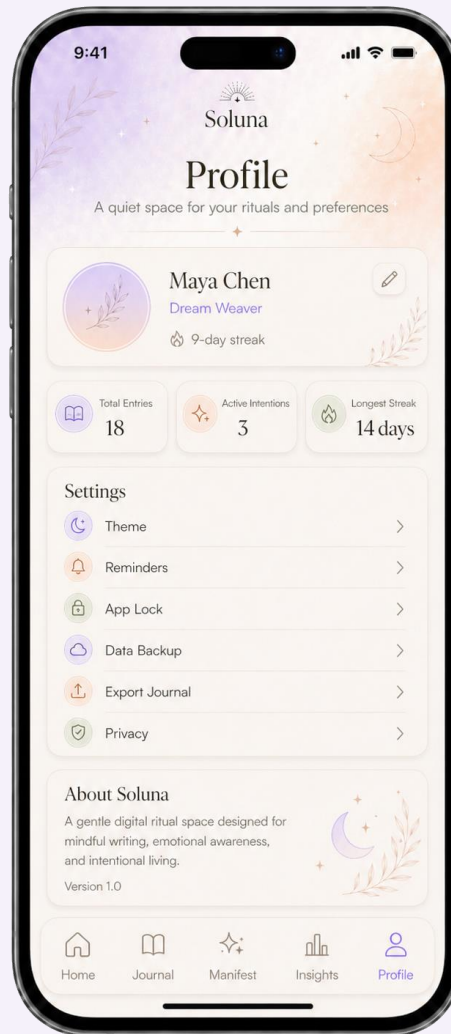
Screen 7. New Manifest Creation Page

Brief function: Guides users to create a new intention using structured input fields.



Screen 8. Insights Page

Brief function: Shows mood calendar, trend graph, habit progress, and monthly summary.



Screen 9. Profile Page

Brief function: Combines profile, settings, personal stats, and About information.

2.2 Consistency of Colour Palette, Typography, and Alignment

The UI screens are designed as a unified system. The following checklist explains how to maintain consistency between screens.

Design Aspect	Consistency Method
① Color Palette	Warm cream backgrounds are used as the base, while lavender, peach, sage green, and dusty pink are used as soft accent colours.
② Typography	Elegant serif headings are used for page titles, while clean modern text is used for labels, buttons, and body content.
③ Component Style	Cards, buttons, chips, input fields, and navigation bars share rounded corners, thin borders, and soft shadows.
④ Alignment	Screens use consistent margins, centred headers, balanced card widths, and clear vertical spacing.
⑤ Navigation	The main app pages use the same bottom navigation: Home, Journal, Manifest, Insights, and Profile.
⑥ Visual Identity	Moon, star, leaf, and gradient details are repeated subtly to support the calm digital ritual theme.

The checklist above summarises the main methods used to maintain the consistency of the UI screens of Soluna. In order to further illustrate this point, the following design system overview (**Figure 1**) intuitively shows the common colour palette, typography, component style, iconography and layout rules in the whole application. This helps to show that each screen is not designed independently, but a unified and coherent visual system.

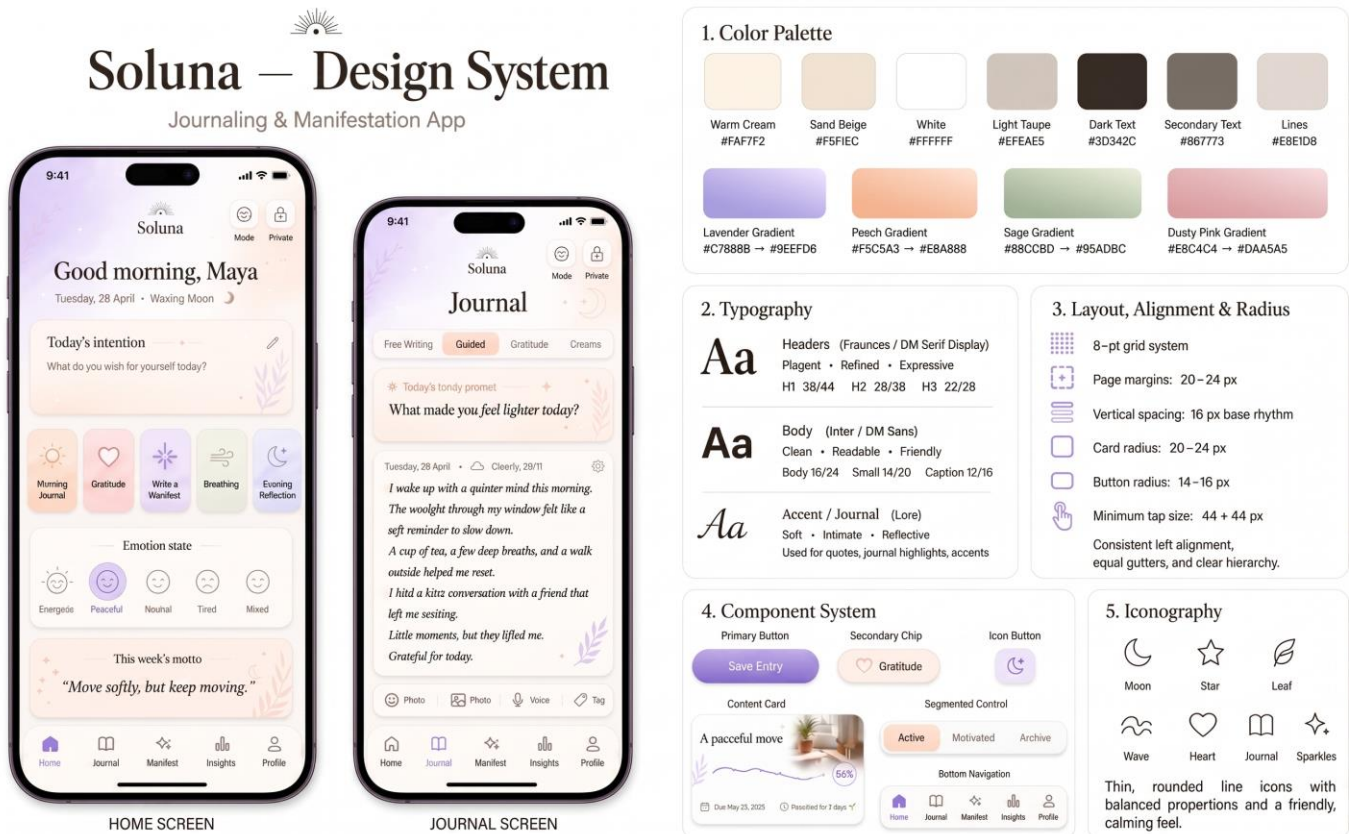


Figure 1. Design system overview of Soluna.

3. Task 3: Navigation and Programming Concepts

3.1 Navigation Structure Diagram

Figure 2 shows the navigation structure of Soluna. The application mainly adopts two navigation methods: **stack navigation** and **bottom tab navigation**. Stack navigation is used at the beginning of the user flow, and when the user enters the main part of the application, the bottom tab navigation is used.

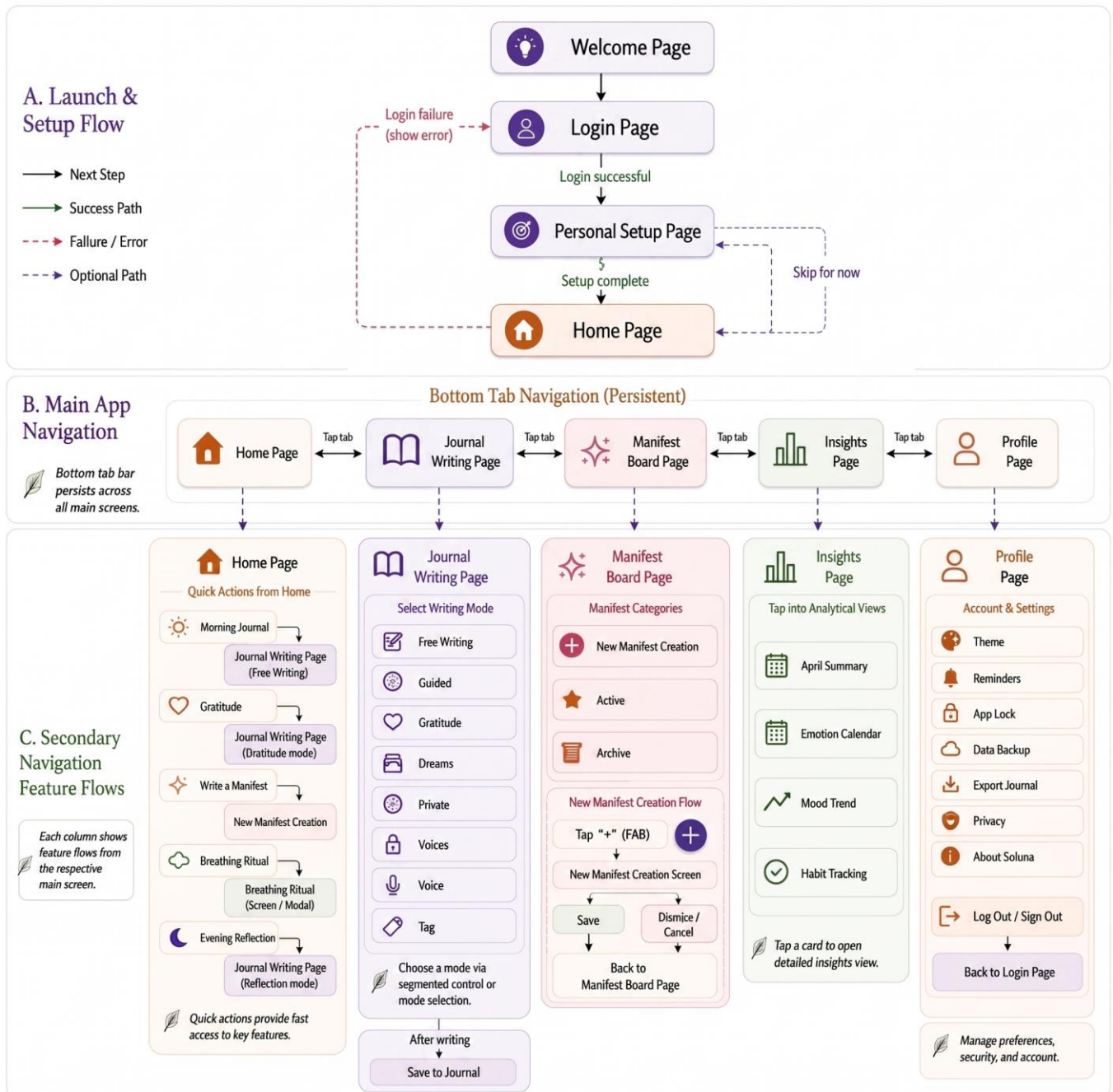


Figure 2. Navigation structure of the Soluna mobile application.

The main structure of the app can be summarized as follows:

- Welcome Page → Login Page → Personal Setup Page → Home Page
- After entering the Home Page, users can browse in the main interface of the application through the bottom tab navigation.
- The five main sections are: Home Page, Journal Writing Page, Manifest Board Page, Insights Page, and Profile Page.
- Each main page connects to related functions, such as quick actions, journal modes, manifest creation, insights modules, and settings options.

3.2 Brief Explanation of Screen Transitions

The focus here is not on writing the actual code, but on explaining the programming concept behind the screen flow. In Soluna, screen transitions can be understood through the following concepts.

Stack Navigation

The working principle of stack navigation is similar to a pile of screens. When the user moves forward, the new page will be superimposed on the current page. For example, the user enters the Login Page from the Welcome Page, then enters the Personal Setup Page, and finally reaches the Home Page. This method is suitable for the user-guided process, because users can operate along a clear step-by-step path.

Bottom Tab Navigation

After entering the main application, users can switch between the main sections through the bottom navigation bar. When the user clicks on a label, the application will directly display the selected section without restarting the whole process. For example, clicking "Journal" will open the Journal Writing Page, and clicking "Insights" will open the Insight Page. This method is very suitable for my Soluna, because its main page needs frequent visits.

Action-based Navigation

Some screens are opened after the user taps a specific button or feature card. For example, tapping "Write a Manifest" opens the Manifest Board Page, and tapping the create button on the Manifest Board Page opens the New Manifest Creation page. This allows the app to connect specific user actions with the correct destination screen.

Return Transition

After completing a task, the application can bring the user back to the previous page. For example, after saving a new manifest, the application will return to the Manifest Board Page. From the perspective of programming, this is equivalent to removing the current page from the navigation stack and displaying the previous page again.

Passing Data Between Screens

When moving from one page to another, the application may pass the selected information to the next screen. For example, you can pass the selected journal mode, the selected target or the newly created manifest content so that the correct information can be displayed on the next page.

In general, Soluna uses stack navigation for the onboarding flow, bottom tab navigation for the main app sections, and action-based navigation for specific feature flows.

4. Task 4: Reflection and Future Improvement

Through this design process, I learned that mobile application design is not only to make the interface visually more attractive, but also to ensure that every design decision can meet the needs of users. The research of Nielsen Norman Group also confirms this, which points out that visual design can not only make the interface look beautiful, but also improve ease of use, stimulate emotions and strengthen brand awareness ([Nielsen Norman Group, 2022](#)). At first, I focused more on Soluna's aesthetic style. For example, I have considered using a stronger lavender background, similar to #E6DAF8, because it looked more decorative and matched the manifestation theme. However, after placing several UI screens together, I found that the purple tone made the report page look slightly crowded and reduced the calm feeling of the app. Therefore, I changed the main background to a warmer cream tone, such as #FAF7F2, and used lavender colour mainly as an accent colour. This makes the application look softer and more suitable for diary recording.

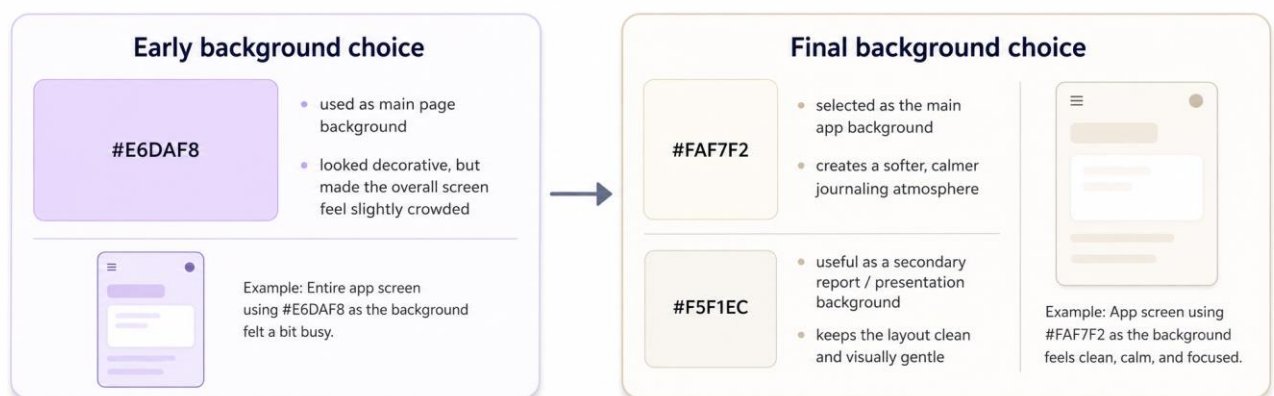


Figure 3. Optimised for the background colour selection of Soluna.

In my future project, I will optimise the design through simple user testing. For example, I can ask friends to complete tasks such as writing a journal or creating a manifest goal, and then observe where they will hesitate. This will help me find out the real useability problems, instead of judging the design from my own perspective.

Reference

- Apple. (2026). *Tab bars*. Apple Human Interface Guidelines. <https://developer.apple.com/design/human-interface-guidelines/tab-bars>
- Faudzi, M. A., Cob, Z. C., Ghazali, M., Omar, R., & Sharudin, S. A. (2024). User interface design in mobile learning applications: Developing and evaluating a questionnaire for measuring learners' extraneous cognitive load. *Heliyon*, 10(18), e37494. <https://doi.org/10.1016/j.heliyon.2024.e37494>
- Fialkowski, B., & Schofield, D. (2024). Considering color: Applying psychology to improve the use of color in digital interfaces. *Art and Design Review*, 12, 306–329. <https://doi.org/10.4236/adr.2024.124022>
- Google. (n.d.). *Navigation bar*. Material Design 3. <https://m3.material.io/components/navigation-bar/overview>
- Gosetto, L., Ehrler, F., Falquet, G., Lovis, C., & McLoughlin, S. (2023). Personalization of mobile apps for health behavior change: Protocol for a cross-sectional study. *JMIR Research Protocols*, 12, e38603. <https://doi.org/10.2196/38603>
- Laubheimer, P. (2016, November 6). *Cards: UI-component definition*. Nielsen Norman Group. <https://www.nngroup.com/articles/cards-component/>
- Nielsen, J. (2006, December 3). *Progressive disclosure*. Nielsen Norman Group. <https://www.nngroup.com/articles/progressive-disclosure/>
- Nielsen Norman Group. (2022, November 6). *Visual design in UX: Study guide*. <https://www.nngroup.com/articles/visual-design-in-ux-study-guide/>

MARKING RUBRICS

Criteria	Poor (0-2)	Average (3-5)	Good (6-8)	Excellent (9-10)	Weight (%)	Mark
App Concept & Creativity (Task 1)	Idea unclear, lacks relevance or originality.	Basic idea stated but limited creativity or user focus.	Clear and relevant idea with some innovative elements.	Strong, creative concept aligned with current mobile design trends and user needs.	10	
	Poor (0-2)	Average (3-5)	Good (6-8)	Excellent (9-10)		
UI Layout & Design Quality (Task 2)	Unclear layout, inconsistent color/font usage, poor spacing.	Layout is understandable but lacks balance and polish.	Organized layout with good color contrast and usability.	Professional, visually cohesive layout with excellent balance and intuitive design.	10	
	Poor (0-2)	Average (3-5)	Good (6-8)	Excellent (9-10)		
Navigation Flow & Logic (Task 3)	No or incomplete navigation flow; unclear screen linkage.	Basic flow shown but lacks logical connections or clarity.	Functional and logical navigation flow with some transitions described.	Well-structured, clearly explained navigation with effective screen relationships.	10	
	Poor (0-2)	Average (3-5)	Good (6-8)	Excellent (9-10)		
Reflection & Continuous Learning (Task 4)	No reflection or poor understanding of design improvement.	Minimal reflection with general comments only.	Clear and thoughtful reflection with self-assessment.	Insightful reflection showing awareness of learning, improvement, and user-centred design.	10	
Total						/40

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